

DAX & Data Visualization

Using Power BI

Project Objective:

The objective of this project is to analyze sales performance across different categories ,sub-categories, cities,and states using Power BI. The dashboard helps stakeholders monitor sales, profit, targets, quantity sold and order distribution to identify business trends and make data-driven decisions.

Dataset Description:

This dataset contains transactional sales data used to evaluate business performance across products, categories, locations and time periods. The data supports analysis of sales trends, target achievement, profitability, customer purchasing behavior and regional performance, multiple related tables were integrated in power BI to create an interactive dashboard using data modeling and DAX calculations.

The dataset enables:

- Monitoring overall Sales, Profit and Order Volume
- Comparing Actual Sales vs Target Sales
- Identifying high-performing categories and sub-categories
- Tracking monthly sales trends
- Analyzing geographical sales distribution
- Measuring profit margins and business performance indicators

Dataset Tables:

1. **List of Orders** - Contains order details and customer location information.
2. **Order Details** - Includes product sales transaction s and quantities.
3. **Sales Target** - Stores target values for performance comparison.
4. **Order Profit** - Contains profit values for profitability analysis.

Key Business Metrics Generated:

- Total Sales
- Total Profit
- Total Order s
- Profit Margin (%)

- Target Achievement (%)
- Monthly Sales Trend
- Category & Sub-category performance
- City/State-wise Sales Distribution

Column Description:

Column Name	Description
Order ID	Unique identifier for each order
Order Date	Date of purchase
Customer Name	Name of the customer
State	State where the order was placed
City	City where the order was placed
Category	Product category (Electronics, Clothing, Furniture)
Sub-Category	Product sub-category
Quantity	Number of products sold
Amount / Sales	Revenue generated from order
Profit	Profit earned from the sale
Profit Margin	Percentage of profit over sales
Target Sales	Expected sales target
Target	Sales target assigned
Month	Sales month for trend analysis

Data cleaning & transformation:

Data Cleaning:

- Imported datasets into Power BI Power Query Editor.
- Checked and removed duplicate records.
- Identified and handled missing/null values.
- Corrected data types for columns (Text, Number, Date, Currency)

- Renamed columns for better readability and consistency.
- Removed unnecessary columns to improve model performance.
- Standardized category and sub-category values to avoid inconsistencies.

Data Transformation:

- Merged related tables using **Order ID** and category fields.
- Created relationships between:
 - **List of Orders** → **Order Details**
 - **Order Details** → **Sales Target**
 - **Order Details** → **Order Profit**
- Created calculated measures using **DAX**, including:
 - **Total Sales** = SUM(Amount)
 - **Total Profit** = SUM(Profit)
 - **Profit Margin** = DIVIDE(Total Profit, Total Sales, 0)
 - **Target Achievement %**
- Generated date-based fields for **monthly sales trend analysis**.
- Created KPIs to compare **Actual Sales vs Target Sales**.
- Applied aggregations and filters for interactive reporting.
- Built hierarchies for **Category** → **Sub-Category** and **State** → **City** analysis.

Visualization and Insights

(Attach Screenshot of Visualizations)

1. KPI Cards

1. Monthly Sales and Total Sales by Profit



2. Minimum Target and Count of Order ID by Target



3. Total Target and Total Sales by Sub-Category



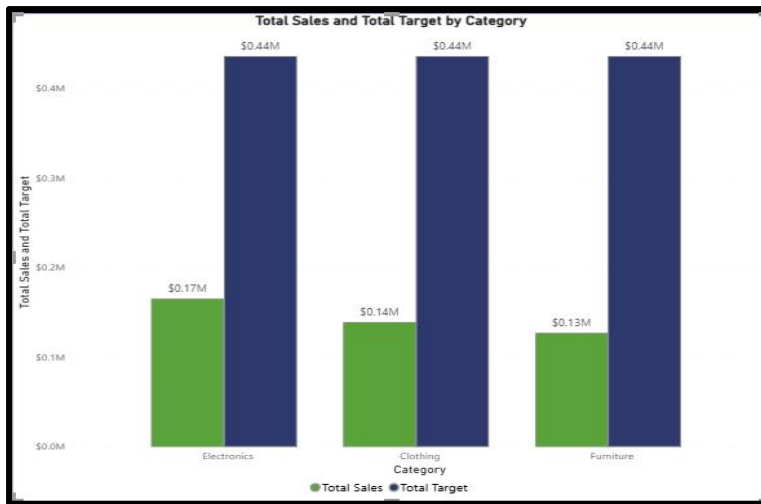
2. Matrix / Table Visual

1. Quantity, Clothing, Electronics, Furniture, Total

Quantity	Clothing	Electronics	Furniture	Total
1	\$174,000	\$129,000	\$132,900	\$435,900
2	\$174,000	\$129,000	\$132,900	\$435,900
3	\$174,000	\$129,000	\$132,900	\$435,900
4	\$174,000	\$129,000	\$132,900	\$435,900
5	\$174,000	\$129,000	\$132,900	\$435,900
6	\$174,000	\$129,000	\$132,900	\$435,900
7	\$174,000	\$129,000	\$132,900	\$435,900
8	\$174,000	\$129,000	\$132,900	\$435,900
Total	\$174,000	\$129,000	\$132,900	\$435,900

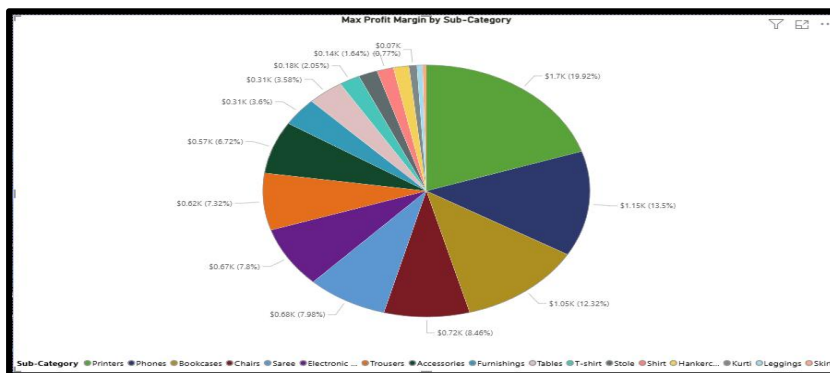
3. Clustered Column Chart

1. Total Sales and Total Target by Category



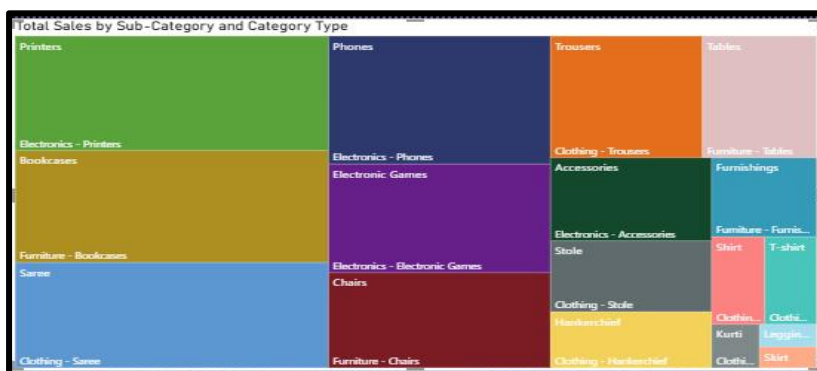
4. Pie Chart

1. Max Profit Margin by Sub-Category



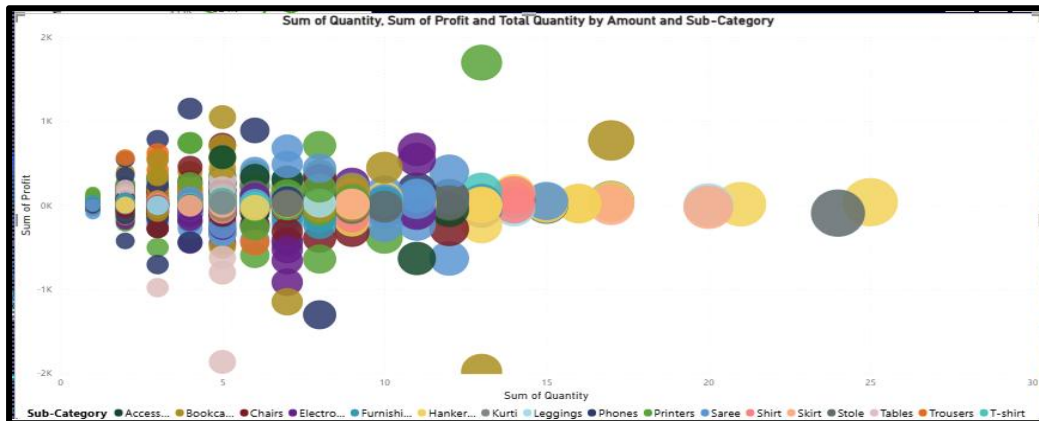
5. Treemap

1. Total Sales by Sub-Category and Category Type



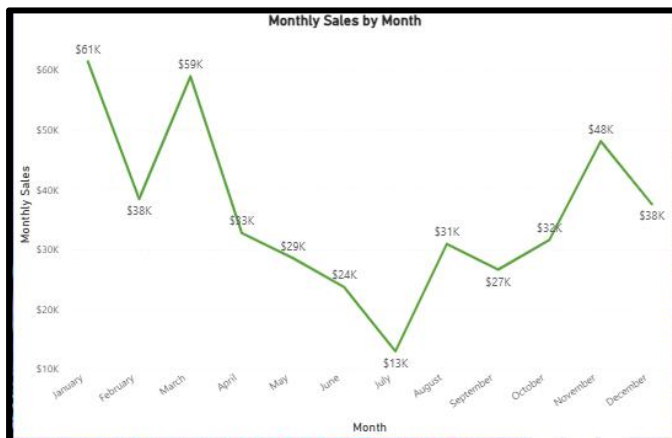
6. Scatter Chart

1. Sum of Quantity, Sum of Profit and Total Quantity Amount by Sub-Category



7. Line Chart

1. Monthly Sales by Month



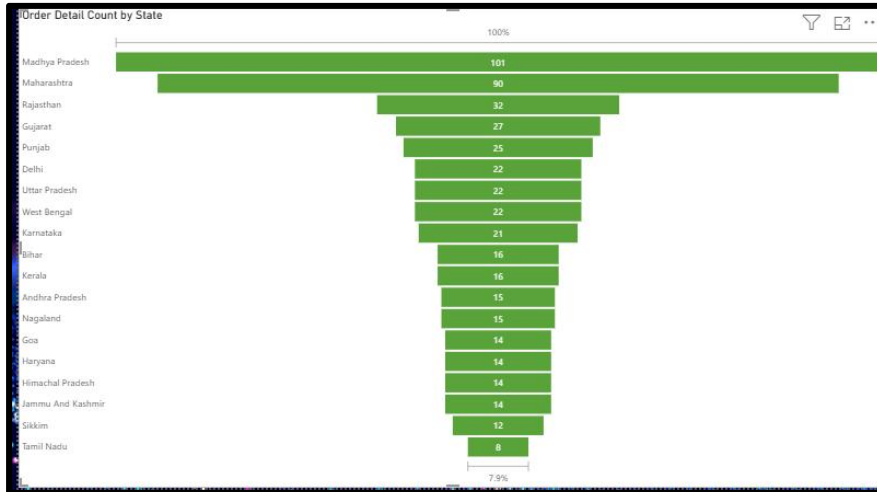
8. Map Visual

1. City Sales by City



9. Funnel Chart

1. Order Detail Count by State



Report / Dashboard Screenshot

This dashboard combines:

- **Descriptive Analysis** → Sales, targets, quantity, profit overview
- **Diagnostic Analysis** → Compare category performance and profit contribution
- **Geographical Analysis** → City-wise sales distribution
- **Trend Analysis** → Monthly sales movement



Analysis

• Descriptive Analysis

Purpose:

Summarize historical business performance.

Findings:

- Total Sales achieved approximately **\$435.9K**
- Target achieved exceeded expectation in several categories.
- Electronics generated the highest sales contribution.
- Monthly sales showed fluctuations across periods.
- Some sub-categories contributed significantly more profit.

Recommendation:

- Continue investment in top-performing categories.

- Monitor low-performing months.

Action Plan:

- ✓ Track monthly KPI dashboard
- ✓ Review category contribution monthly
- ✓ Create executive summary reports

• Diagnostic Analysis

Purpose:

Identify root causes behind performance trends.

Findings:

- Electronics exceeded target due to stronger sales volume.
- Certain sub-categories generated high sales but lower profit margins.
- Geographic distribution indicates concentrated sales regions.

Recommendation:

- Investigate low-profit categories.
- Optimize pricing and promotional strategy.

Action Plan:

- ✓ Conduct category-level profitability review
- ✓ Improve inventory planning
- ✓ Segment customer demand by region

• Predictive Analysis

Purpose:

Forecast future sales and business performance.

Findings:

- Monthly sales trend indicates seasonal fluctuations.
- High-performing categories may continue growth if demand remains stable.
- Current trajectory suggests target achievement is sustainable.

Recommendation:

- Implement sales forecasting models.
- Prepare stock for high-demand categories.

Action Plan:

- ✓ Build forecasting model in Power BI
- ✓ Set automated alerts for declining sales
- ✓ Predict next-quarter demand

● Prescriptive Analysis

Purpose:

Recommend actions to improve future outcomes.

Findings:

- Increase focus on profitable categories.
- Reduce dependency on underperforming products.
- Improve regional sales strategies.

Recommendation:

- Introduce targeted campaigns.
- Optimize pricing and promotions.
- Improve supply chain responsiveness.

Action Plan:

- ✓ Allocate budget to top categories
- ✓ Launch category-specific campaigns
- ✓ Set quarterly target monitoring process

Conclusion – Sales Report Dashboard

The Sales Dashboard provides a comprehensive view of overall business performance across sales, targets, profit, categories, and geographic distribution.

- **Sales Performance:** Total sales reached approximately **\$435.9K**, indicating strong business activity and performance close to or above several target benchmarks.
- **Target Achievement:** Monthly and category-level comparisons show that actual sales performed well against defined targets, with some areas exceeding expectations.
- **Category Insights:** **Electronics** generated the highest sales contribution, followed by **Clothing** and **Furniture**, making Electronics the strongest revenue-driving category.
- **Profit Analysis:** Profit distribution across sub-categories shows that a few product groups contribute significantly more profit than others, highlighting opportunities to focus on high-margin segments.
- **Sub-Category Trends:** Categories such as **Printers, Phones, and Bookcases** contributed noticeably to total sales, while some sub-categories showed lower performance and may require strategic attention.
- **Geographical Performance:** City-wise sales analysis indicates stronger sales concentration in selected regions, suggesting opportunities for targeted regional expansion.
- **Order & Quantity Patterns:** Order count and quantity analysis reveal variations in customer purchasing behavior across categories and locations.

1. LIST OF ORDER

Order ID	Order Date	CustomerName	State	City	Location
B-25627	Monday, April 23, 2018	Hitika	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25637	Thursday, April 26, 2018	Ashmi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25645	Tuesday, May 1, 2018	Yaanvi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25651	Monday, May 7, 2018	Anurag	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25655	Friday, May 11, 2018	Nida	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25663	Saturday, May 19, 2018	Pournamasi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25680	Monday, June 4, 2018	Aayushi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25681	Monday, June 4, 2018	Bhawna	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25685	Sunday, June 10, 2018	Sheetal	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25688	Monday, June 11, 2018	Swetha	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25690	Friday, June 15, 2018	Gunjan	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25692	Sunday, June 17, 2018	Rashmi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25701	Tuesday, June 26, 2018	Maithilee	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25703	Thursday, June 28, 2018	Ekta	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25705	Saturday, June 30, 2018	Shweta	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25708	Sunday, July 1, 2018	Kishwar	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25709	Sunday, July 1, 2018	Aakanksha	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25719	Thursday, July 12, 2018	Rashmi	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25724	Thursday, July 19, 2018	Sheetal	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25727	Sunday, July 22, 2018	Turumella	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25734	Sunday, July 29, 2018	Pranav	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25737	Wednesday, August 1, 2018	Shubham	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25741	Friday, August 3, 2018	Navdeep	Madhya Pradesh	Indore	Indore,Madhya Pradesh
B-25746	Tuesday, August 14, 2018	Divyanshu	Madhya Pradesh	Indore	Indore,Madhya Pradesh

2. ORDER - PROFIT

File	Home	Help	Table tools	Share	
Name	Order - Profit	Manage relationships	New measure	Quick measure	New column
Structure	Category	Average Profit	New table	Mark as date table	Calendars
	Furniture	9.45679012345681			
	Clothing	11.7629083245522			
	Electronics	34.0714285714286			

3. ORDER DATA

File	Home	Help	Table tools	Share	
Name	Order Data	Manage relationships	New measure	Quick measure	New column
Structure	Order ID	Order Date	CustomerName	State	City
	B-26081	Friday, March 22, 2019	Aarushi	Tamil Nadu	Chennai
	B-26018	Thursday, February 14, 2019	Aarushi	Tamil Nadu	Chennai
	B-26008	Saturday, February 9, 2019	Kalyani	Tamil Nadu	Chennai
	B-25860	Thursday, November 15, 2018	Akshay	Tamil Nadu	Chennai
	B-25788	Friday, September 21, 2018	Dinesh	Tamil Nadu	Chennai
	B-25716	Wednesday, July 11, 2018	Surabhi	Tamil Nadu	Chennai
	B-25698	Saturday, June 23, 2018	Amisha	Tamil Nadu	Chennai
	B-25608	Sunday, April 8, 2018	Aarushi	Tamil Nadu	Chennai

4. ORDER DETAILS

File	Home	Help	Table tools	Share	
Name	Order Details	Manage relationships	New measure	Quick measure	New column
Structure	Order ID	Amount	Profit	Quantity	Category
	B-25603	12	\$1	2	Clothing
	B-25608	257	\$23	5	Clothing
	B-25615	68	\$20	5	Clothing
	B-25616	42	\$12	5	Clothing
	B-25624	26	\$12	3	Clothing
	B-25625	97	\$29	2	Clothing
	B-25635	40	\$16	3	Clothing
	B-25638	154	\$39	3	Clothing
	B-25654	34	\$12	3	Clothing
	B-25656	6	\$3	1	Clothing
	B-25656	56	\$18	2	Clothing
	B-25670	24	\$1	2	Clothing
	B-25670	14	\$2	1	Clothing
	B-25730	18	\$8	2	Clothing
	B-25751	32	\$7	3	Clothing
	B-25757	106	\$15	7	Clothing
	B-25757	14	\$5	1	Clothing
	B-25757	17	\$7	3	Clothing
	B-25771	148	\$59	3	Clothing
	B-25783	26	\$2	2	Clothing
	B-25784	15	\$4	1	Clothing
	B-25793	18	\$1	3	Clothing
	B-25797	31	\$1	2	Clothing
	B-25800	45	\$12	7	Clothing

5. ORDER DETAILS - SUMMARY

File Home Help Table tools Share

Name Order Details - Su...

Manage relationships Relationships

New measure Quick measure New column New table Mark as date table Calendars

Structure

Order ID	Order Count
B-25605	1
B-25606	1
B-25607	1
B-25611	1
B-25612	1
B-25613	1
B-25615	1
B-25617	1
B-25619	1
B-25620	1
B-25622	1
B-25624	1
B-25627	1
B-25629	1
B-25631	1
B-25632	1
B-25634	1
B-25636	1
B-25637	1
B-25639	1
B-25641	1
B-25642	1
B-25644	1
B-25646	1

Table: Order Details - Summary (500 rows)

Data

Search

- List of Orders
- Order - Profit
- Order Data
- Order Details
- Order Details - Summary
- Sales target
- Sales target - Summary
- Total Amount

6. SALES TARGET

File Home Help Table tools Share

Name Sales target

Manage relationships Relationships

New measure Quick measure New column New table Mark as date table Calendars

Structure

Month of Order Date	Category	Target	Target Achievement
Saturday, April 18, 2026	Furniture	10400	1222.89%
Monday, May 18, 2026	Furniture	10500	1211.25%
Thursday, June 18, 2026	Furniture	10600	1199.82%
Saturday, July 18, 2026	Furniture	10800	1177.60%
Tuesday, August 18, 2026	Furniture	10900	1166.80%
Friday, September 18, 2026	Furniture	11000	1156.19%
Sunday, October 18, 2026	Furniture	11100	1145.77%
Wednesday, November 18, 2026	Furniture	11300	1125.50%
Friday, December 18, 2026	Furniture	11400	1115.62%
Monday, January 19, 2026	Furniture	11500	1105.92%
Thursday, February 19, 2026	Furniture	11600	1096.39%
Thursday, March 19, 2026	Furniture	11800	1077.81%
Saturday, April 18, 2026	Clothing	12000	1158.78%
Monday, May 18, 2026	Clothing	12000	1158.78%
Thursday, June 18, 2026	Clothing	12000	1158.78%
Saturday, July 18, 2026	Clothing	14000	993.24%
Tuesday, August 18, 2026	Clothing	14000	993.24%
Friday, September 18, 2026	Clothing	14000	993.24%
Sunday, October 18, 2026	Clothing	16000	869.09%
Wednesday, November 18, 2026	Clothing	16000	869.09%
Friday, December 18, 2026	Clothing	16000	869.09%
Monday, January 19, 2026	Clothing	16000	869.09%
Thursday, February 19, 2026	Clothing	16000	869.09%
Thursday, March 19, 2026	Clothing	16000	869.09%

Table: Sales target (36 rows)

Data

Search

- List of Orders
- Order - Profit
- Order Data
- Order Details
- Order Details - Summary
- Sales target
- Sales target - Summary
- Total Amount

7. SALES TARGET - SUMMARY

File Home Help **Table tools** Share

Name: Sales target - Sum...

Manage relationships | New measure | Quick measure | New column | New table | Mark as date table

Structure

Month	Total Target
April	31400
May	31500
June	31600
July	33800
August	33900
September	34000
October	36100
November	36300
December	36400
January	43500
February	43600
March	43800

Data

- List of Orders
- Order - Profit
- Order Data
- Order Details
- Order Details - Summary
- Sales target
- Sales target - Summary**
- Total Amount

8. TOTAL AMOUNT

File Home Help **Table tools** Share

Name: Total Amount

Manage relationships | New measure | Quick measure | New column | New table | Mark as date table

Structure

Sub-Category	Total Amount
Bookcases	56861
Stole	18546
Hankerchief	14608
Electronic Games	39168
Phones	46119
Saree	53511
Trousers	30039
Chairs	34222
Kurti	3361
T-shirt	7382
Shirt	7555
Leggings	2106
Tables	22614
Printers	58252
Accessories	21728
Furnishings	13484
Skirt	1946

Data

- List of Orders
- Order - Profit
- Order Data
- Order Details
- Order Details - Summary
- Sales target
- Sales target - Summary
- Total Amount**